



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PHYSICS
PAPER 2

6032/2
1 hour 30 min

JUNE 2025 SESSION

Additional materials:
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

For numerical answers, **all** working should be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
6	
TOTAL	

- 1 (a) Define *impulse* and state its SI unit.

_____ [2]

- (b) (i) State *Stoke's Law*.

_____ [1]

- (ii) A raindrop of radius, r , and density, ρ , falls through a fluid of viscosity, η , and density, σ , with a terminal velocity, v . The viscosity is given by the equation, $\eta = \frac{2gr^2(\rho - \sigma)}{9v}$.

1. Determine the base units for η .

2. Calculate the terminal velocity of the raindrop.

[diameter of raindrop is 3.0 cm, viscosity of air is $1 \times 10^{-3} \text{ Nm}^{-2}\text{s}$,
density of water is 1000 kgm^{-3} , density of air is 1 kgm^{-3}]

[3]

- (c) Give any **two** applications of *Newton's 3rd law of motion*.

[2]

- 2 (a) State the *principle of superposition*.

[1]

- (b) (i) Explain the term

1. *node*,

[1]

2. *antinode.*

[1]

(ii) Give **two** differences between *progressive* and *stationary waves*.

[2]

(c) The velocity, v , at time, t , of an oscillating particle is defined by the equation,
 $v = 3.18 \sin 1.59t$.

(i) Determine the

1. amplitude of the particle,

2. antinode.

[1]

(ii) Give **two** differences between *progressive* and *stationary waves*.

[2]

(c) The velocity, v , at time, t , of an oscillating particle is defined by the equation,
 $v = 3.18 \sin 1.59t$.

(i) Determine the

1. amplitude of the particle,

2. maximum acceleration of the particle.

[1]

- (ii) Sketch a graph showing how the acceleration varies with displacement for the particle in (i).

[2]

- 3 (a) (i) Define *the tesla*.

[1]



- (ii) An aluminum power cable carries a current of 1000 A between two poles 50 m apart. The magnetic flux density due to the earth is about $35 \mu\text{T}$ at an angle of 60° to the horizontal.

Calculate the

1. magnetic force per unit length,

[2]

2. resistance of the wire,
{diameter of the wire is 2.5 cm and the resistivity of aluminum is $2.63 \times 10^{-8} \Omega\text{m}$ }

[2]

3. electrical power dissipated by the wire.

[1]



- (b) (i) Give **two** properties of an ideal op-amp.

_____ [2]

- (ii) State any **two** input transducers.

_____ [2]

- 4 (a) The pressure, P , of a fixed mass of an ideal gas is related to its temperature, T , by the equation

$$P = \alpha kT$$

where k is the Boltzmann's constant.

State the

- (i) significance of α ,

_____ [1]

- (ii) units of α ,

_____ [1]



- (iii) A fixed mass of a gas has a pressure, P , temperature, T , and volume, V . When temperature is decreased to $0.25T$, pressure decreases to $0.5P$.

Determine the final kinetic energy of the gas in terms of P and V .

[3]

- (b) Fig 4.1 shows a circular pipe of radius 3.10 cm at A and 6.00 cm at B. A liquid of mass 3.5 kg and density 0.70 g/cm^3 passes through point B per minute.

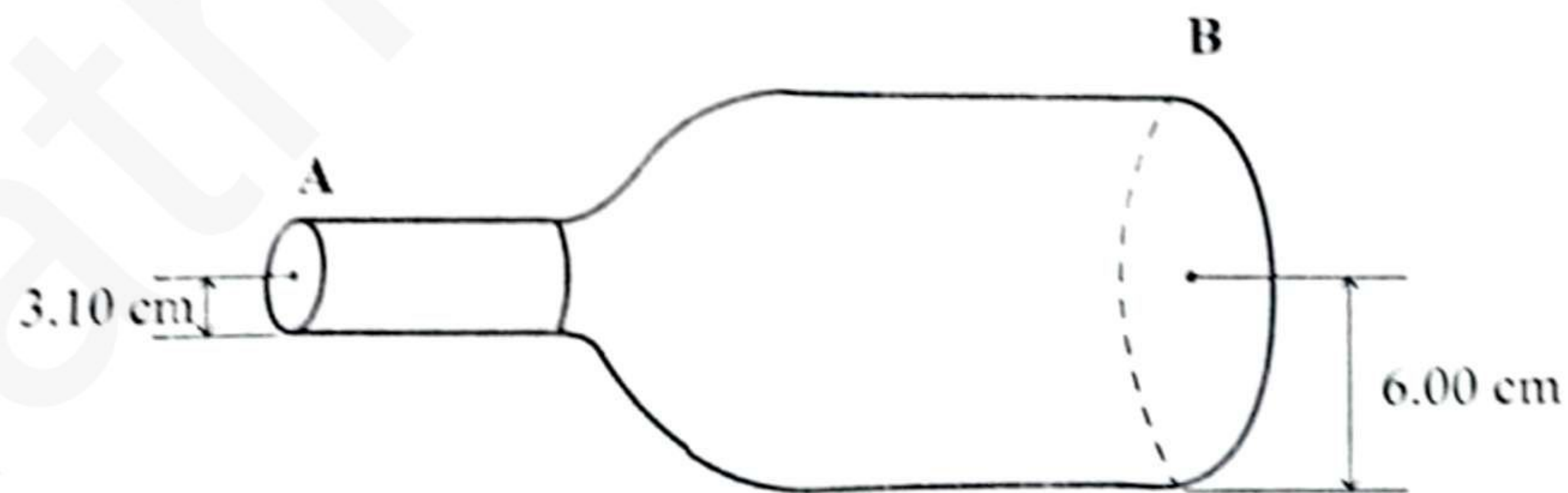


Fig 4.1

6032 2 J2025



Calculate the

(i) volume flow rate,

[2]

(ii) pressure difference between A and B.

[3]



- 5 (a) **Fig 5.1** shows apparatus for the measurement of specific charge of an electron e/m . The electron gun fires electrons at a speed of $3.6 \times 10^6 \text{ ms}^{-1}$ normally into a uniform magnetic field.

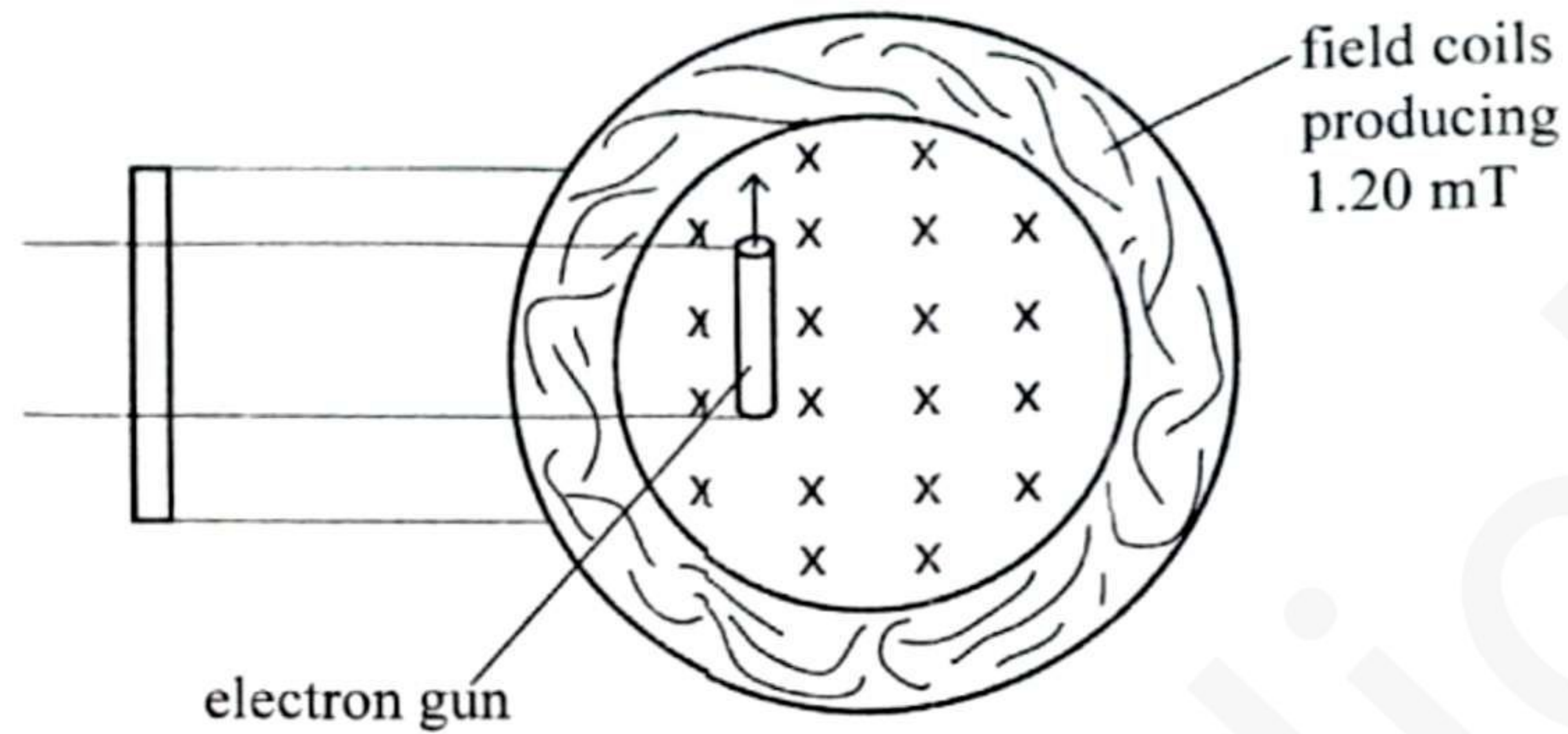


Fig 5.1

- (i) On **Fig 5.1**, complete the path followed by electrons. [1]
- (ii) Derive the equation for determining e/m .

- (iii) The electrons follow a path of diameter 34 mm. Find the value of [1]

1. specific charge, e/m ,

[2]



2. mass of an electron, m_e .

For
Examiner's
Use

[2]

- (b) (i) Define the term *half life*.

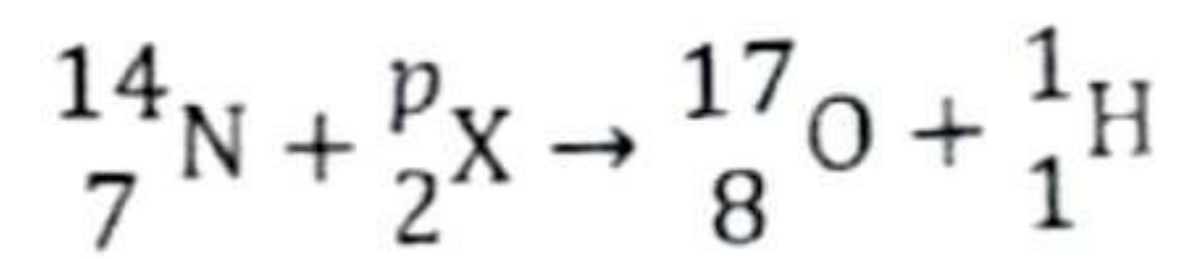
[1]

- (ii) A radioactive sample irradiates 8000 nuclei per second and contains 6.2×10^{14} atoms.
Calculate the half life of the sample.

[3]



- 6 (a) Nitrogen and another atom X react to form an isotope of oxygen in the nuclear process:



mass of ${}^{17}_8\text{O} = 16.99913\text{u}$

mass of proton = 1.00728u

mass of neutron = 1.00867u

mass of electron = 0.00055u

- (i) Determine the value of p .

[1]

- (ii) Calculate the binding energy per nucleon of ${}^{17}_8\text{O}$.

[3]

- (iii) Predict the stability of ${}^{17}_8\text{O}$.

[2]



- (b) (i) State the effects of increasing the following on the production of an input signal

1. sampling rate,

[1]

2. number of bits.

[1]

- (ii) The attenuation of an information signal along a conductor is 45 dB.
The initial information signal has a power of 60 mW.

Calculate the output power.

[2]

